Available online at www.sciencedirect.com
SCIENCE DIRECT®
Combustion and Flame 144 (2006) 205–224
Combustion and Flame
www.elsevier.com/locate/combustflame

Investigations of swirl flames in a gas turbine model combustor I. Flow field, structures, temperature, and species distributions

P. Weigand, W. Meier *, X.R. Duan [1], W. Stricker, M. Aigner

Institut für Verbrennungstechnik, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Pfaffenwaldring 38, D-70569 Stuttgart, Germany

Received 22 November 2004; received in revised form 2 June 2005; accepted 8 July 2005

Available online 21 September 2005

Abstract

A gas turbine model combustor for swirling CH_4/air diffusion flames at atmospheric pressure with good optical access for detailed laser measurements is discussed. Three flames with thermal powers between 7.6 and 34.9 kW and overall equivalence ratios between 0.55 and 0.75 were investigated. These behave differently with respect to combustion instabilities: Flame A burned stably, flame B exhibited pronounced thermoacoustic oscillations, and flame C, operated near the lean extinction limit, was subject to sudden liftoff with partial extinction and reanchoring. One aim of the studies was a detailed experimental characterization of flame behavior to better understand the underlying physical and chemical processes leading to instabilities. The second goal of the work was the establishment of a comprehensive database that can be used for validation and improvement of numerical combustion models. The flow field was measured by laser Doppler velocimetry, the flame structures were visualized by planar laser-induced fluorescence (PLIF) of OH and CH radicals, and the major species concentrations, temperature, and mixture fraction were determined by laser Raman scattering. The flow fields of the three flames were quite similar, with high velocities in the region of the injected gases, a pronounced inner recirculation zone, and an outer recirculation zone with low velocities. The flames were not attached to the fuel nozzle and thus were partially premixed before ignition. The near field of the flames was characterized by fast mixing and considerable finite-rate chemistry effects. CH PLIF images revealed that the reaction zones were thin ($\leqslant$0.5 mm) and strongly corrugated and that the flame zones were short ($h \leqslant 50$ mm). Despite the similar flow fields of the three flames, the oscillating flame B was flatter and opened more widely than the others. In the current article, the flow field, structures, and mean and rms values of the temperature, mixture fraction, and species concentrations are discussed. Turbulence intensities, mixing, heat release, and reaction progress are addressed. In a second article, the turbulence–chemistry interactions in the three flames are treated.
© 2005 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

* Corresponding author. Fax: +49 711 6862 578.
E-mail address: wolfgang.meier@dlr.de (W. Meier).
[1] Present address: Southwestern Institute of Physics, P.O. Box 432, 610041 Chengdu Sichuan, People's Republic of China.

0010-2180/$ – see front matter © 2005 The Combustion Institute. Published by Elsevier Inc. All rights reserved.
doi:10.1016/j.combustflame.2005.07.010

Keywords: Gas turbine; Model combustor; Swirl flame; Turbulence–chemistry interaction; Validation measurements; Laser techniques

1. Introduction

Swirl flames are used extensively in practical combustion systems because they enable high energy conversion in a small volume and exhibit good ignition and stabilization behavior over a wide operating range [1–4]. In stationary gas turbine (GT) combustors, they are used mostly as premixed or partially premixed flames, and in aero engines, as diffusion flames. To reduce pollutant emissions, especially NO_x, today the flames are operated generally very lean [5–7]. Under these conditions, the flames tend to exhibit undesired instabilities, e.g., in the form of unsteady flame stabilization or thermoacoustic oscillations. The underlying mechanisms of the instabilities are based on the complex interaction between flow field, pressure, mixing, and chemical reactions, and are not well enough understood to date. Detailed measurements in full-scale combustors are hardly possible, and very expensive and numerical tools have not yet reached a sufficient level of confidence to solve the problems. A promising strategy lies therefore in the establishment of a laboratory-scale "standard combustor" with practical relevance and detailed, comprehensive measurements using nonintrusive techniques with high accuracy. The gained data set will be used for validation and optimization of numerical combustion simulation codes which then can be applied to simulate the behavior of technical combustors. Intrusive probe measurements are less suited for these applications as they disturb the local flow field and change the conditions for stabilization and for reaction—locally or even in general [8,9]. In turbulent reacting flows, the use of optical measurement techniques is therefore essential for reliable information. Laser-based tools are the method of choice offering the potential to measure most of the important quantities with high temporal and spatial resolution, often as one- or two-dimensional images, and the ability to perform the simultaneous detection of several quantities [10–13]. The flow field can be measured by laser Doppler velocimetry (LDV) or particle imaging velocimetry (PIV); the temperature by laser Rayleigh scattering, laser Raman scattering, coherent anti-Stokes Raman scattering (CARS), or laser-induced fluorescence (LIF); species concentrations by LIF or Raman scattering; flame structures by planar LIF (PLIF) or PIV; and the mixture fraction by Raman scattering.

In recent years a variety of laser-based investigations in GT model combustors have been reported that, besides feasibility studies, concentrated on certain aspects of the combustion process or model validation. For example, Kaaling et al. [14] performed temperature measurements with CARS in a RQL (rich-quench-lean) combustor, and Kampmann et al. [15] used CARS simultaneously with 2-D Rayleigh scattering to characterize the temperature distribution in a double-cone burner. In the same combustor, Dinkelacker et al. [16] studied the flame front structures with PLIF of OH and 2-D Rayleigh scattering. Fink et al. [17] investigated the influence of pressure on the combustion process by applying PLIF of OH and NO in a LPP (lean prevaporized premixed) model combustor. With respect to NO_x reduction strategies, Cooper and Laurendeau [18,19] performed quantitative NO LIF measurements in a lean direct-injection spray flame at elevated pressures. Shih et al. [20] applied PLIF of OH and seeded acetone in a lean premixed GT model combustor, and Deguchi et al. [21] used PLIF of OH and NO in a large practical GT combustor. Hedman and Warren [22] used PLIF of OH, CARS, and LDV for the characterization of a GT-like combustor fired with propane in order to achieve a better understanding of the fundamentals of GT combustion. PLIF of OH was also applied by Lee et al. [23] to study flame structures and instabilities in a lean premixed GT combustor, by Arnold et al. [24] to visualize flame fronts in a GT combustor flame of 400 kW, and by Fritz et al. [25] for revealing details of flashback. Löfström et al. [26] performed a feasibility study of two-photon LIF of CO and 2-D temperature mapping by LIF of seeded indium in a low-emission GT combustor. A comparison of two different laser excitation schemes for major species concentration measurements with laser Raman scattering was performed by Gittins et al. [27] in a GT combustion simulator. At a high-pressure test rig of the DLR, various laser techniques (LDV, CARS, PLIF of OH and kerosene, and 2-D temperature imaging via OH PLIF) were applied to GT combustors under technical operating conditions to achieve a better understanding of combustor behavior and to validate CFD codes [28–31].

In the study discussed here, a nozzle with two concentric air swirlers and an annular fuel supply between them was used for CH_4/air diffusion flames, with thermal powers up to 35 kW at atmospheric pressure. The combustion chamber enabled almost unrestricted optical access to the flames and was, thus, ideally suited for the application of laser measurement techniques. The velocity fields were mea-

sured by 3-D LDV, the flame structures by PLIF of OH and CH, and the joint probability density functions (PDFs) of temperature, major species concentrations, and mixture fraction by laser Raman scattering. Three flames with different characteristics were investigated: flame A was operated at a specific power rate of 42.4 MW/(m^3 bar) that is comparable to the values of aeronautical or modern aero-derivative industrial gas turbines, which are operated around 25 to 70 MW/(m^3 bar); flame B was chosen at a power rate of 12.5 MW/(m^3 bar) that is comparable to most industrial gas turbines, which are operated at 5 to 20 MW/(m^3 bar); and flame C was operated at the same airflow as flame B but with reduced fuel supply close to the lean extinction limit (with a power rate of 9.2 MW/(m^3 bar)). This is of interest because modern gas turbines in power plants are operated under extremely lean conditions to meet the emission limits. In addition, the flames were operated at three different equivalence ratios to investigate the stabilization of the flames. Flame A with an equivalence ratio of $\Phi = 0.65$ burned stably, whereas flame B ($\Phi = 0.75$) emitted strong thermoacoustic noise, and flame C with $\Phi = 0.55$ operated close to the blowoff limit and randomly experienced sudden liftoff and reestablishment of stable operation.

The advantage of the combustor setup used was the excellent optical access to the flame zone, enabling the collection of information from the whole area around the flame zone in a burner that is close to technical application. In particular, the detailed velocity measurements at the nozzle exit result in well-defined boundary conditions, which are important for numerical methods. One major goal of the work was the detailed experimental analysis of the flames to gain deeper insight into, e.g., the mixing and stabilization processes, the shape of the reaction zones and the regions of heat release, and effects of turbulence–chemistry interactions. The second goal was the establishment of a comprehensive database which can be used for the verification and improvement of combustion simulation codes. The present article focuses on flow fields, on the distribution of the temperature, the major species concentrations and the mixture fractions, and on the instantaneous and mean flame structures. The turbulence–chemistry interactions, which play an important role in these flames, are discussed in a second article [32]. The thermoacoustic oscillations of flame B were analyzed previously by phase-resolved measurements [33–35]. The results from those investigations represent a supplement of the measurements without phase resolution presented in the current article, and some of the findings are used here to support the discussion of the characteristics of flame B.

2. Experimental

2.1. Combustor and flames

The gas turbine model combustor is schematically shown in Fig. 1. The burner was a modified version of a practical gas turbine combustor with an air blast nozzle for liquid fuels [36]. Co-swirling dry air at room temperature was supplied to the flame through a central nozzle (diameter 15 mm) and an annular nozzle (i.d. 17 mm, o.d. 25 mm contoured to an outer diameter of 40 mm). Both air flows were fed from a common plenum with an inner diameter of 79 mm and a height of 65 mm. The radial swirlers consisted of 8 channels for the central nozzle and 12 channels for the annular nozzle. The ratio of the air mass flows through the annular and central nozzle was approximately 1.5. Nonswirling CH_4 was fed through 72 channels (0.5 × 0.5 mm), forming a ring between the air nozzles. Compared with an annular nozzle for CH_4 with a slit width of <0.5 mm, this configuration ensured better realization of the cylindrical symmetry of the fuel injection and set well-defined boundary conditions for numerical simulation. The exit planes of the fuel and central air nozzles were located 4.5 mm below the exit plane of the outer air nozzle; the latter was defined as reference height $h = 0$. The combustion chamber had a square section of 85 × 85 mm and a height of 114 mm and consisted of four quartz plates held by steel posts (diam 10 mm) in the corners, thus allowing very good optical access

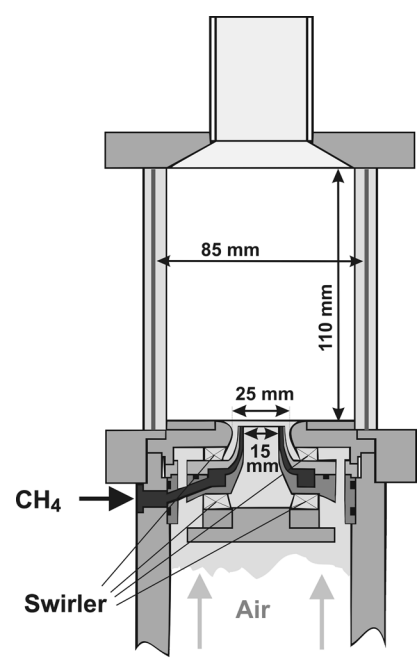

Fig. 1. Schematic drawing of the model combustor.

Table 1
Parameters of the three flames investigated

	Air		CH_4		P_{th}[a] (kW)	Φ_{glob}	f_{glob}	$T_{glob\ ad}$ (K)
	sl/min	g/min	sl/min	g/min				
A	850	1095	58.2	41.8	34.9	0.65	0.037	1750
B	218	281	17.2	12.3	10.3	0.75	0.042	1915
C	218	281	12.6	9.0	7.6	0.55	0.031	1570

[a] P_{th}, thermal power; Φ_{glob}, equivalence ratio for the overall mixture; f_{glob}, mixture fraction for the overall mixture; $T_{glob\ ad}$, adiabatic temperature for the overall mixture with inlet temperature $T_0 = 295$ K.

to the flame. A conical top plate made of steel with a central exhaust tube (diam 40 mm, length 50 mm) formed the exhaust gas exit. The high velocity in the exhaust tube avoided any backflow from outside the combustion chamber.

The three flames investigated were: flame A, with $P_{th} = 34.9$ kW and an overall equivalence ratio of $\Phi_{glob} = 0.65$ that ran very stably; flame B, with $P_{th} = 10.3$ kW and $\Phi_{glob} = 0.75$, which exhibited pronounced self-excited thermoacoustic oscillations at a very high noise level; and flame C, operated close to the lean extinction limit, with $P_{th} = 7.6$ kW and $\Phi_{glob} = 0.55$, which randomly lifted off and reanchored to the normal stabilization height. Table 1 lists characteristic parameters of the investigated flames, i.e., the volume and mass flow rates of air and fuel as well as the resulting values for power and overall "global" values for equivalence ratio, mixture fraction, and adiabatic temperature (given the subscript "glob"). The mass flows of the gases were controlled with Brooks flow controllers (Type 5853S for air and Type 5851S for CH_4) with an accuracy of typically ±0.5%. As can be seen from Table 1, flames B and C had almost identical total flow rates but different flame parameters. To achieve this, the air mass flow was kept constant for both flames and only the fuel mass flow, which represents 7.3% of the total flow for flame B and 5.5% for flame C, was changed. This was chosen to achieve a high similarity of the velocity distributions to exclude flow field effects as a source for the different behavior of the two flames.

All three flames—established under globally lean conditions—showed no soot production and burned with a light blue color. The flames appeared as type 2 swirl flames [1,37] with a conically shaped toroidal flame zone at different opening angles and, as was confirmed by the velocity measurements, showed pronounced recirculation zones on the axis (inner recirculation zone, irz) and near the walls of the combustion chamber (outer recirculation zone, orz). The visual appearance of the flames, despite the square combustion chamber, revealed good rotational symmetry.

The swirl number S was calculated from the velocity profile just above the nozzle exit neglecting the pressure term according to

$$S = \frac{\int_0^R 2\pi u w \rho r \, \mathrm{d}r}{R \int_0^R 2\pi u^2 \rho r \, \mathrm{d}r},$$

where u = axial velocity (m/s), w = circumferential velocity (m/s), ρ = density (kg/m^3), r = radius (m), and R = maximum radius of the nozzle exit (m). The swirl numbers are $S \approx 0.9$ for flame A and $S \approx 0.55$ for flames B and C. Given the fact that the nozzle was contoured and a combustion chamber was used with an expansion factor $D/d = 3.4$ (D = diameter of combustion chamber, d = diameter of nozzle), vortex breakdown with establishment of an irz was to be expected [38]. Due to the confinement, an orz was also found. The nozzle Reynolds number based on the cold inflow and the minimum outer nozzle diameter (25 mm) was about 15,000 for flames B and C and about 58,000 for flame A.

All three flames did not burn directly at the fuel nozzle exit, but rather with a liftoff height of some millimeters. In flame C, sudden liftoff (partial extinction) and flashback randomly occurred (approximately 10 times per minute), as it was chosen to operate close to the lean extinction limit that was found to be at $\Phi = 0.53$. The random liftoff which reached up to a height of 30–40 mm lasted typically 100–150 ms. Thus, flame C was, for about 2% of the time, in the partially extinguished mode. Of the three flames discussed here, flame B exhibited the highest noise level with a quite discrete frequency of about 290 Hz. Flame C showed nearly the same frequency spectrum as flame B but with a much reduced amplitude. Flame A emitted a rather broadband noise with small peak amplitudes at about 380 Hz.

2.2. Measuring techniques

All three velocity components were measured simultaneously using commercial LDV systems (DISA/DANTEC) and a cw-Ar$^+$ laser (Coherent, INNOVA 90, operated at 1 W). The optical arrangement consisted of a two-component system (DISA 55X, $\lambda_{laser} = 488$ and 514.5 nm) and a single-component system (DISA Flow Direction Adapter, $\lambda_{laser} =$

514.5 nm), which were arranged orthogonally and both used in the forward scatter mode. The laser beams were transmitted via mono mode fibers into the optical modules. A frequency shift of 40 MHz was applied for all three directions; the center frequency of the detection was adapted to each measuring point. Focusing lenses with $f = 300$ mm were used; the resulting probe volumes were about 60 µm in diameter and 1.0 mm in length for x and y directions, and 120 µm in diameter and 1.5 mm in length for the z direction, corresponding to the axial (u), radial (v), and tangential (w) directions of the velocity, respectively. Because the spatial intensity distribution of the Mie scattering with its maximum in forward direction is more than 100 times higher than at 90°, the two systems could be operated at the same wavelength (here 514.5 nm) because the forward-scattered signals can be clearly discriminated by their intensities in this orthogonal geometry. The detection optics were arranged in the y–z plane at about 10° off axis and consisted of commercial camera lenses ($f = 85$ mm, f/1.8 for x and y directions and $f = 105$ mm, f/4 for the z direction) focusing the signals on photomultiplier tubes. For the simultaneous detection and analysis of the photomultiplier signals, three Dantec burst spectrum analyzers (BSA enhanced, 57N20 and 57N35) were used with a record length of 64 points. ZrO_2 particles with a diameter of approximately 2 µm were seeded as scatterers into the airflow. Because of the small size, the particles can follow flow field fluctuations up to a frequency of 1.2 kHz within an accuracy of 99%. At each measuring location, typically 10,000 to 15,000 validated velocity data were recorded, except in the area of the flame zone, where sometimes only 2000 samples were validated during the record time. For flames B and C, measurements at lower heights were carried out coincidentally with a time filter of 2 µs, thus providing also the Reynolds stress tensors and cross moments. Using only the coincidental values, the effective probe volume and, consequently, the data rate are drastically reduced, leading to much longer acquisition time. Therefore, this was done only at selected heights. For calculating mean values, the noncoincidental values were used for an improved statistic. The lowest height for LDV measurements was $h = 1.5$ mm for flame A. In flames B and C an improvement of the setup enabled measurements as low as $h = 1.0$ mm. For simplicity in this study, these levels are all labeled $h = 1$ mm.

Planar laser-induced fluorescence (PLIF) of OH and CH radicals was applied to visualize the flame structures. A Nd:YAG laser pumped optical parametric oscillator (Spectra Physics GCR 290 and MOPO 730) was used to supply pulsed laser radiation for the excitation of OH and CH radicals. The laser beam was formed to a light sheet ($h \approx 45$ mm) and irradiated vertically into the flame intersecting the flame axis. The pulse energies were typically 3 mJ/pulse for OH and 4 mJ/pulse for CH with a bandwidth of about 0.45 cm^{-1}. The sheet thickness was approximately 0.25 mm in the imaged area. The resulting spectral laser intensities are on the order of 16 $MW/cm^2\,cm^{-1}$ for CH and 12 $MW/cm^2\,cm^{-1}$ for OH. Compared with the saturation intensities, which are around 1 $MW/cm^2\,cm^{-1}$ for the chosen transitions [39,40], the applied laser intensities are relatively high and a significant degree of saturation is expected. The excited fluorescence was collected at 90° by a lens (for OH: achromatic UV lens, $f = 100$ mm, f/2, Halle Nachf.; for CH: camera lens, $f = 50$ mm, f/0.95, Canon) and, after spectral filtering, detected with an intensified CCD camera (for OH: LaVision Flamestar II; for CH: Roper Scientific). The laser pulse duration was 5 ns; the temporal detection gate of the image intensifier was 50 ns for OH and 200 ns for CH (limited by irising effects of the image intensifier). OH radicals were excited on the $R_1(8)$ line of the $A^2\Sigma^+$–$X^2\Pi$ ($\nu' = 1$, $\nu'' = 0$) transition at $\lambda = 281.3$ nm [41] and the fluorescence was detected through an interference filter in the wavelength region $\lambda \approx 312 \pm 10$ nm. For CH, the $Q_1(7)$ line of the $B^2\Sigma$–$X^2\Pi$ ($\nu' = 0$, $\nu'' = 0$) band was excited at 390.3 nm [42,43]. For suppression of laser-scattered light and background radiation, a filter combination of a KV418 (Schott) and a short-pass filter with a cutoff wavelength of 450 nm (Oriel) was used in front of the camera, and only the fluorescence in the B–X $(0, 1)$ and A–X bands around $\lambda \approx 430$ nm was detected. The A state is efficiently populated by collision-induced electronic energy transfer from the B to the A electronic level [44,45].

For pointwise quantitative measurement of the concentrations of major species (O_2, N_2, CH_4, H_2, CO, CO_2, H_2O) and temperature, laser Raman scattering was used [46]. The radiation of a flashlamp-pumped dye laser (Candela LFDL 20, wavelength $\lambda = 489$ nm, pulse energy $E_p \approx 3$ J, pulse duration $\tau_p \approx 3$ µs) was focused into the combustion chamber, and the Raman scattering emitted from the measuring volume (length ≈0.6 mm, diam ≈0.6 mm) was collected by an achromatic lens ($D = 80$ mm, $f = 160$ mm) and relayed to the entrance slit of a spectrograph (SPEX 1802, $f = 1$ m, slit width 2 mm, dispersion 0.5 nm/mm). The dispersed and spatially separated signals from the different species were detected by photomultiplier tubes in the exit plane of the spectrograph and sampled by boxcar integrators. The species number densities were calculated from these signals using calibration measurements, and the temperature was deduced from the total number density via the ideal gas law [46,47]. The simultaneous detection of all major species with each laser pulse also

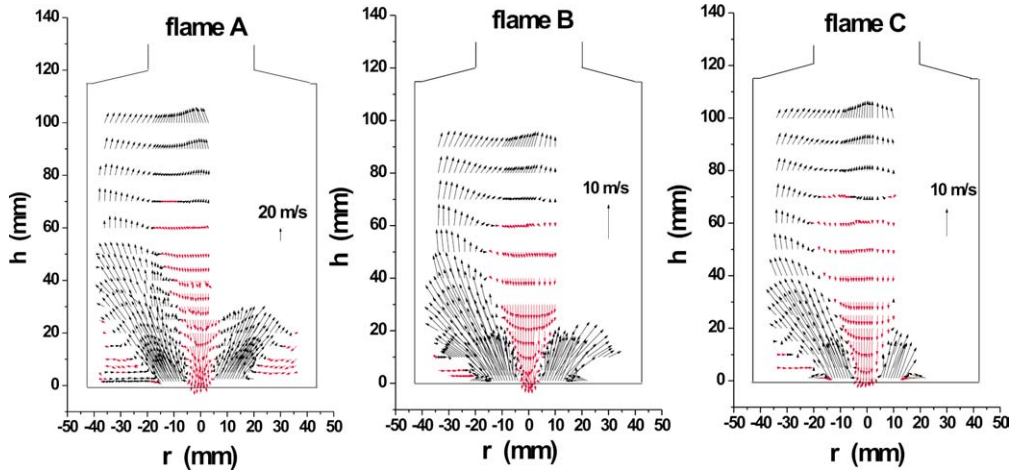

Fig. 2. Vectorplots of combined uv velocities for flames A, B, and C; negative u velocities are displayed in red. The lines indicate the size of the combustion chamber.

enabled the determination of the instantaneous mixture fraction [48]. At each measuring location 500 single-pulse measurements were performed within a scanning pattern of roughly 100 points, from which the joint probability density functions (PDFs) were computed. The choice of 500 samples turned out to be a good trade-off between measuring time and convergence of the mean and rms values. Studies in highly turbulent regions of these flames revealed that the "final" values are reached to within 2% after 300 to 400 samples. The measurement uncertainty for the mean values of temperature, mixture fraction, and mole fraction of O_2, H_2O, and CO_2 is typically 3–4%.

3. Results and discussion

3.1. LDV measurements

As expected for this type of confined swirl flame, the mean flow field of each of the three flames shows a strong inner recirculation zone (irz) along the axial centerline as well as an outer recirculation zone (orz) near the walls of the combustion chamber, as can be seen in the vectorplots of the combined uv velocities displayed in Fig. 2. The inlet velocities at the lowest measuring height $h = 1$ mm correspond to the mass flows, with the maximum values for the axial velocity of $u_{max} = 39.0$ m/s in flame A, $u_{max} = 13.3$ m/s in flame B, and $u_{max} = 13.0$ m/s in flame C. The angle of the maximum mean velocity in the uv plane with respect to the axial centerline is about 26° for all three flames for $h = 10$–20 mm. The measurements also revealed (not shown) that the ratio of the tangential velocity w and the axial velocity u is nearly double for flame A ($S \approx 0.9$) compared with flames B and C ($S \approx 0.55$). Nevertheless, the resulting flow fields are very similar, despite the different visible appearance of the three flames. The normalized mean axial velocities (u_{mean}/u_{max}) at $h = 1$ mm, illustrated in Fig. 3, are almost identical and show that the inflow at this height extends radially from $r = 5$ to 15 mm. The isolines of $u_{mean} = 0$, plotted in Fig. 4, indicate the boundaries of the irz. It can be seen that for flames A and C, the irz reaches up to $h \approx 73$ mm, whereas in flame B it ends at $h \approx 62$ mm. In the near field of the nozzle, for $h < 10$ mm, the contours of the irz are nearly identical for all three flames, and in all three flames the irz extends below the lowest measuring level. It can therefore be assumed that the irz reaches even into the central air nozzle, which ends at $h = -4.5$ mm. In comparing the mean values of the different flames, it must, however, be considered that flame B is subject to periodic oscillations and that the time-averaged velocities represent an average not only over turbulent fluctuations but also over periodic variations (see discussion below).

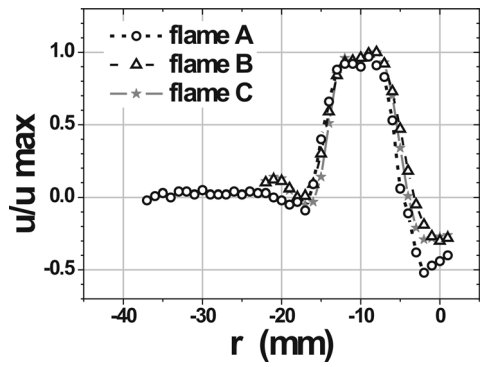

Fig. 3. Radial profiles of the normalized mean axial velocity (u/u_{max}) at $h = 1$ mm for flames A, B, and C.

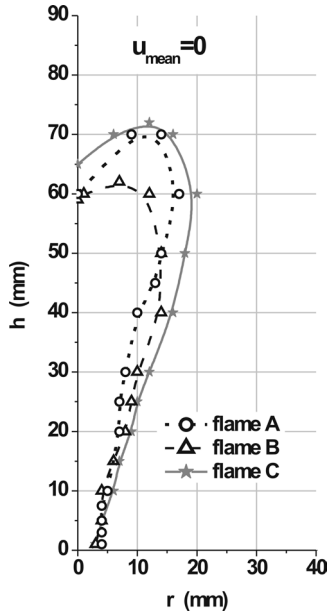

Fig. 4. Isolines of $u_{\text{mean}} = 0$ representing the extension of irz for flames A, B, and C.

Fig. 5 illustrates the mean values and rms fluctuations of the three velocity components of flame A for $h = 1.5$, 5, 15, and 45 mm. At $h = 1.5$ mm, the profile of the axial velocity u reflects the inflow of the fresh gas at $r \approx 5$–16 mm with maximum values around 39 m/s and the irz with velocities of $u \approx -20$ m/s. The radial velocity component v is negative for $r >$ 16 mm, reflecting the size of the orz. In the inflow, v is roughly half as large as u. The tangential velocity w is rather constant in the orz ($w \approx 10$ m/s), and its radial profile displays two maxima in the region of the inflow, which likely reflect the flows from the two air nozzles with the minimum between them originating from the fuel nozzle and its wake. For $r = 0$–5 mm, w increases linearly with r, reflecting the solid body rotation part of the vortex. The rms values of u and v have a pronounced maximum in the shear layer between the inflow and the irz, and v exhibits another maximum in the shear layer between the inflow and the orz. The high level of the rms values close to the flame axis demonstrates that the flow field is subject to strong turbulent fluctuations in this region of the flame. At $h = 5$ mm, the gradients of the radial profiles of the mean and rms values have become smaller in comparison to $h = 1.5$ mm, but the basic features of the flow are unchanged. In the orz, u_{mean} is close to zero but u_{rms} is 6–9 m/s. This shows that u changes frequently its direction in the orz. At $h = 15$ mm, the profiles have broadened and the reverse flow on the axis reaches its highest negative velocity of $u_{\text{mean}} \approx -26$ m/s. The orz has shrunk but is still discernible from the negative v component. The mean tangential velocity component indicates a solid body vortex up to $r \approx 10$ mm. For $r > 10$ mm, w_{mean} declines but the shape of the radial profile does not resemble that of a potential vortex. w_{rms} is quite constant over the radius, whereas u_{rms} and v_{rms} exhibit broad maxima. With increasing downstream position, the profiles smooth out and the orz vanishes, whereas the irz reaches a radial expansion of $r \approx 13$ mm at $h = 45$ mm.

In the shear layer between the inflow and the irz, large velocity fluctuations and the low mean velocity generally cause a very high turbulence intensity ($u_{\text{rms}}/u_{\text{mean}} \gg 100\%$). Therefore, intense mixing of the cold fresh gas with hot burned gases coming from the irz can be expected in this region. Fig. 6 shows, for example, radial profiles of normalized u_{rms} at $h = 10$ mm, normalized by the maximum velocities at $h = 1$ mm; i.e., $u_{\text{max}} = 39.0$, 13.3, and 13.0 m/s for flames A, B, and C, respectively. The broad peaks of the u_{rms} values around $r \approx 6$ mm indicate that the instantaneous flow fields are subject to strong turbulent fluctuations and that the shear layer is not locally stable. This finding is also supported by the single shot 2-D LIF images that are discussed in the following paragraphs. In Fig. 6 it can also be seen that for the three flames, the relative velocity fluctuations $u_{\text{rms}}/u_{\text{max}}$ are very similar and the values of u_{rms} reach more than 50% of u_{max} at $r \approx 3$–10 mm. Therefore, the irz should not be regarded as a stable structure with the fluid following streamlines that hardly vary their positions. However, from the measurements performed in this study, there was no indication of coherent structures such as rotating vortex pairs.

To demonstrate that the average flow conditions at the nozzle exit for flames B and C were similar, the radial velocity profiles of u, v, and w at the exit of the nozzle are plotted in Fig. 7. As clearly can be seen, the mean profiles of all three velocity components match very well as intended, so that the mean flow field can be excluded as the primary reason for the different behavior of these two flames. However, periodic variations would not be revealed by the mean profiles. One also recognizes the good symmetry of the time-averaged velocity profiles, as is expected for a rotational symmetric flow. The dips in the profiles that can be seen at $r \approx 10$–12 mm result from the wake of the fuel nozzle and have already disappeared at $h = 5$ mm (not shown) due to the high turbulence. The orz becomes apparent by the inward-directed radial velocity v in the region $r > 15$ mm. Further downstream, the different thermal powers and combustion temperatures of flames B and C lead to a different thermal expansion which has an influence on the flow velocities. To demonstrate this effect, Fig. 8 shows the radial

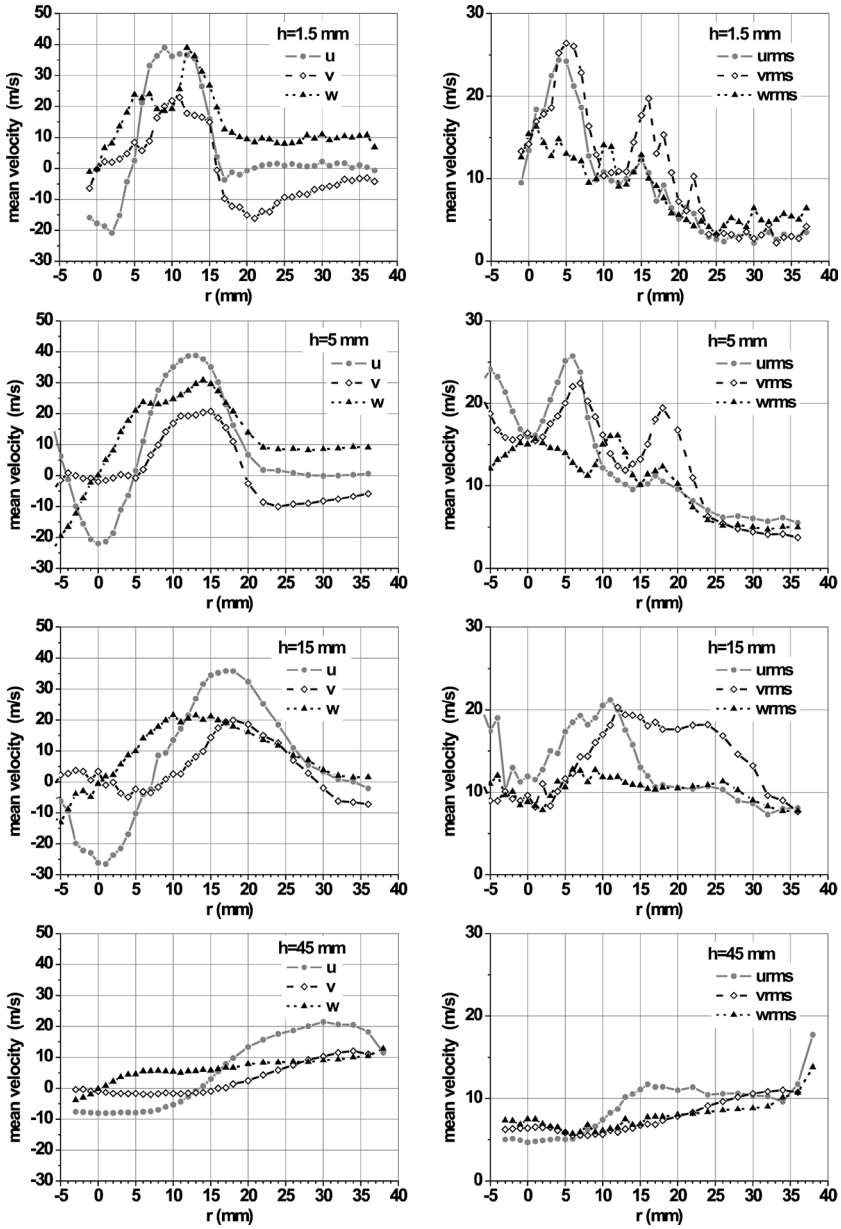

Fig. 5. Radial profiles of the mean values (left side) and rms fluctuations (right side) of the three velocity components in flame A at different heights.

profiles of u, v, and w at $h = 30$ mm. Here, the axial velocity in flame B is significantly larger than in flame C; e.g., $u_{\max}$ is 10.6 m/s in flame B and 8.3 m/s in flame C. The profiles of the radial velocity component are almost identical and those of the tangential velocity show only slightly higher velocities for flame B. Thus, the different thermal expansion of the flames influences predominantly the axial velocity, as expected for confined flames. It is also typical of confined flames that the swirl number decreases with combustion progress due to the axial acceleration.

In flame B, LDV measurements have also been performed with phase resolution at heights $h = 1$ and 5 mm. In those measurements, the phase of the acoustic oscillation was measured by a microphone simultaneously with two velocity components [34,35]. An important finding was that the irz and orz varied in size during an oscillation cycle, resembling a pumping motion: The irz varied mainly in the axial direction, and the orz in the radial direction. Thus, although the mean flow fields of flames B and C look very similar, they are inherently differ-

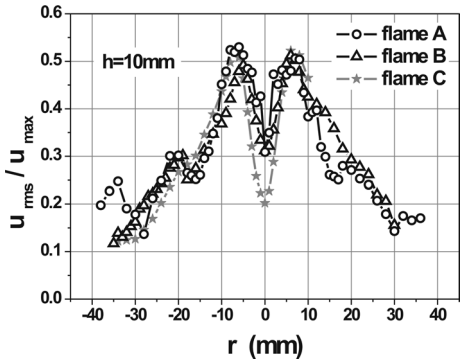

Fig. 6. Radial profiles of u_{rms} at $h = 10$ mm for flames A, B, and C.

ent, with flame B exhibiting periodic variations superposed onto turbulent fluctuations and flames A and C exhibiting only turbulent fluctuations.

3.2. Flame structures from OH LIF and CH LIF measurements

In flames, OH can be found in detectable concentrations at temperatures above approximately 1400 K, especially in fuel-lean mixtures [49]. The equilibrium OH concentration increases exponentially with temperature but differently for fuel-lean and fuel-rich mixtures. Furthermore, OH is formed in superequilibrium concentrations in the reaction zones, and its relaxation to equilibrium by three-body collisions is quite slow at atmospheric pressure ($\tau > 3$ ms) [50]. Thus, high OH LIF intensities can be regarded as an indicator of hot gas and/or reacting fuel/air mixtures. CH radicals are formed at high temperatures on the fuel-rich side of the reaction zone and have a much shorter lifetime ($\approx$10 ns) than OH radicals [51]. Thus, high CH concentrations can be interpreted as a marker for the fuel consumption layer of the reaction zone and, with some restrictions, as a qualitative measure of the heat release rate [52]. For illustration, Fig. 9 shows the calculated profiles of the temperature and OH and CH mole fractions as a function of the mixture fraction f for a strained laminar CH_4/air counterflow diffusion flame with a strain rate of $a = 400\ \mathrm{s}^{-1}$ [53,54]. Significant CH concentrations are present only in mixtures with $f \approx 0.05$–0.08. In contrast, OH is also found in lean mixtures and covers a range of $f \approx 0.015$–0.08. It can also be seen that CH is a factor of about 500 lower in concentration than OH and, thus, much harder to detect. Although the turbulent flames investigated cannot be directly compared with a counterflow diffusion flame, this example shows at least qualitatively the characteristic behavior of OH and CH.

With respect to the interpretation of the LIF images presented here, one has to keep in mind that the LIF intensities are not necessarily proportional to the species number density. The relative population of the initial state excited by the laser changes with temperature (Boltzmann fraction f_{B}). For OH, the Boltzmann fraction of the initial rotational state, $N'' = 8$, varies by less than 7% over the temperature range of interest ($T \approx 1400$–2200 K). For CH, f_{B} of $N'' = 7$ decreases by roughly 20% over the temperature range 1700 to 2200 K. Because the LIF signals are quenching dominated, variations in quenching environment can significantly influence the fluorescence yield. An estimation was performed for OH using the LASKIN program [55] and the temperature and gas composition from the strained laminar flame calculation with $a = 400\ \mathrm{s}^{-1}$ already used in Fig. 9. It turned out that the OH fluorescence yield varied by about 15% over the range of interest. Taking into account that the composition of the flame under investigation may deviate from the strained laminar flame composition, the measured LIF signal intensities reflect the OH density roughly within 25%. For CH, the situation is more complex because quenching in two electronic states, A and B, and predissociation in the B state play a role. However, in the thin layer of the reaction zone where CH is present, the gas composition and temperature do not change drastically and variations in quenching are expected to be small. Rensberger et al. [56] reported that changes in the fluorescence quantum yield after excitation of the $B(v' = 0)$ state in different flames were small and that quenching varied by less than 50%. In the flames investigated here, variations should not be larger.

Fig. 10 shows typical OH single-shot LIF distributions for the three flames. The images display the region $r = -41$–41 mm and $h \approx 0$–47 mm. For temperatures below $\approx$1300–1400 K, OH concentrations were below the detection limit (dark areas). For all three flames, the OH distributions cover broad areas with strongly wrinkled contours and sometimes isolated regions (at least in a 2-D cut). These structures yield a good impression of the turbulent transport and mixing processes within the flames. These images show that the instantaneous flame structures (and very likely also the instantaneous flow fields) are much less uniform as might be assumed from the mean flow field. The steep gradients of OH LIF intensities that frequently occurred may represent either a reaction zone or the boundary between cold and hot fluid. The OH-free regions near the nozzle reflect the inlet flow of mostly unreacted fuel and air, which is directed diagonally upward. The mean contours of these regions can be better seen in the averaged images of Fig. 11 (left). The highest mean OH LIF intensities and thus, within 25% uncertainty, the highest mean OH con-

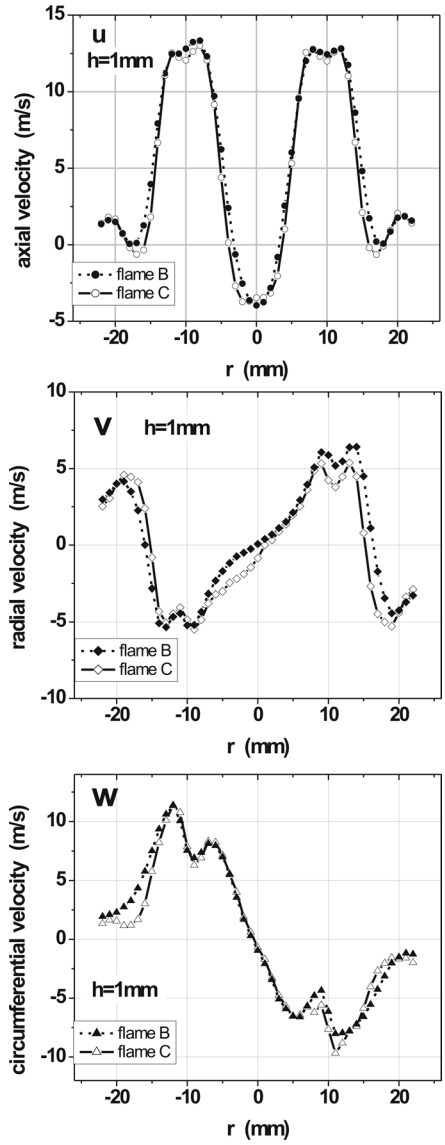

Fig. 7. Radial profiles of the simultaneously measured three velocity components u, v, and w for flames B and C at $h = 1$ mm.

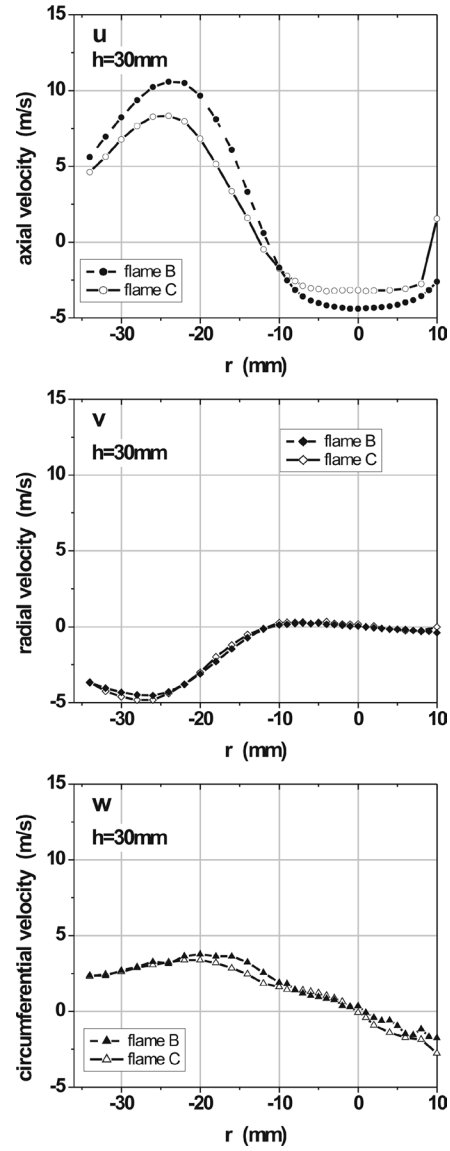

Fig. 8. Radial profiles of the simultaneously measured three velocity components u, v, and w for flames B and C at $h = 30$ mm.

centrations are found in each flame in the irz shortly above the nozzle, with a maximum at $h \approx 10$ mm. Here, flame C has the lowest mean OH concentration, followed by flame B with approximately 50% more and flame A with approximately 75% more. Around $h = 30$ mm on the axis, the concentrations are approximately three times less than at $h = 10$ mm for each flame. These significant OH concentrations in the irz in the averaged images (and also in most of the single shot images) indicate a high temperature in this region. The mixing of this hot gas from the irz with fresh gas from the nozzles presumably plays the key role in the ignition and stabilization of the flames. Investigations in flame B using planar two-line OH LIF thermometry showed that temperature and OH concentrations were not generally well correlated in that flame and that superequilibrium concentrations contributed significantly to the high OH levels within and some millimeters downstream of the reaction zones [57]. Comparison of the OH LIF distributions from the three flames further shows that flame C exhibits a smaller area containing OH, especially in the orz. This is explained by the lower overall temperature level of this flame (see Table 1) accord-

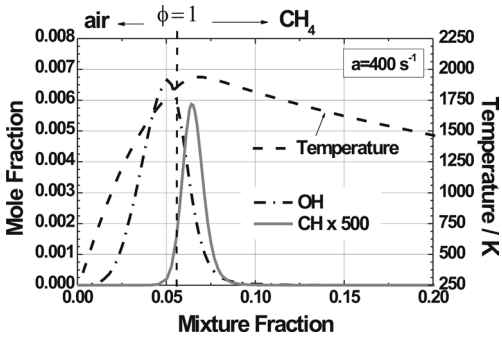

Fig. 9. Calculated profiles for temperature and concentrations of OH and CH radicals in a counterflow diffusion flame with a strain rate of $a = 400\ \mathrm{s}^{-1}$.

ing to the global equivalence ratio. It is also seen that for flame A, the inlet flow of cold gas penetrates further downstream in comparison to the other flames.

The averaged CH LIF distributions displayed in Fig. 11 (right) reflect the regions where flame reactions and heat release take place. The shapes of the three flames are quite different: while flames A and C are conically shaped with opening angles $\theta/2 \approx 30°$ and $\approx 45°$, respectively, flame B has a significantly larger opening angle and is rather flat with $\theta/2 \approx 75°$. The difference between flames B and C is surprising because their mean flow fields are quite similar and the mean velocities are almost identical at $h = 1$ mm (see Fig. 7). Comparing the velocity fields and the CH LIF images it becomes apparent that the heat release for flame C and especially for flame B does not take place predominantly in the shear layer between the irz and the inflow, as might be expected. The CH LIF distribution and the region of the shear layer are in good accordance only for flame A, whereas for flames B and C, the opening angles of the flame zones are larger than that of the shear layer. The unusual behavior of flame B is related to the periodic pulsations and is addressed in Section 3.3. Further, it is important to note that for all three flames the regions of heat release do not begin at the fuel nozzle, but at $h \approx 5$ mm, $h \approx 4$ mm, and $h \approx 6$ mm for flames A, B, and C, respectively. Due to this liftoff, fuel, air, and exhaust gas are already partially premixed before ignition. According to the CH LIF images, the heat release is complete at $h \approx 20$ mm and $h \approx 40$ mm for flames B and C, respectively. In flame A, there is still a small amount of CH at the upper end of the measured area at $h = 47$ mm.

For the interpretation of the averaged distributions one has to keep in mind that flames B and C exhibit unsteady combustion behavior. Flame A is highly turbulent but steady, and mean and rms values are related to turbulent fluctuations. Flame C is subject to sudden partial extinction and can be regarded as bimodal, i.e., either burning stably or not burning up to $h = 30$–40 mm. However, the partially extinguished state is present only about 2% of the time and its contribution to the time-averaged values is small. Thus, flame C should be classified as "nearly steady" and its fluctuations are caused predominantly by turbulence. For flame B, the situation is different, because the thermoacoustic pulsations are permanent and temporal changes of the flame include turbulent fluctuations as well as periodic variations. The mean species and temperature distributions discussed in this article are averaged over both turbulent and periodic changes. To distinguish between them, phase-resolved measurements have to be performed that yield averaged values at distinct phase angles of the periodic pulsation. Such measurements have also been performed for flame B, and the results are discussed in detail in separate publications [33–35].

The single-shot LIF distributions of CH, displayed in Fig. 12, show thin ($\approx$0.3–0.5 mm) and strongly corrugated reaction zones which are, at least in the 2-D cut, sometimes interrupted. Their shapes are dominated by the turbulent flow field and vary strongly from shot to shot. Some samples exhibit only weak flame reactions, while others possess strongly distorted and intense reaction zones. Analysis of a larger number of single shots revealed that in flame A, the CH layers are more intensely contorted and the flame surface area is larger than in flames B and C [58]. Furthermore, CH LIF peak intensities differ in the three flames. They are highest for flame B ($\Phi_{\mathrm{glob}} = 0.75$) followed by flame A ($\Phi_{\mathrm{glob}} = 0.65$) and flame C ($\Phi_{\mathrm{glob}} = 0.55$); i.e., the order is the same as for the global equivalence ratios. This indicates that the reactions occur, on average, at different local equivalence ratios and not generally around $\Phi_{\mathrm{local}} = 1$, as would be assumed for diffusion flames. This observation is confirmed by the Raman results concerning the thermochemical states of the flames.

3.3. Mixture fraction, temperature, and species mole fractions

To yield an overview of the main features of the distributions of mixture fraction f, temperature T, and mole fraction X, the results from the pointwise Raman measurements are displayed as two-dimensional charts which were obtained by interpolating between the measuring locations. With the optical setup of the Raman system, measurements were restricted to the region $h \geqslant 5$ mm and $r \leqslant 30$ mm.

The distributions of the mean mixture fraction, as displayed in Fig. 13, show that the highest f values are found directly above the fuel nozzle exit, as expected. It is, however, remarkable that these values are already quite small at $h = 5$ mm; e.g., $f_{\max} =$

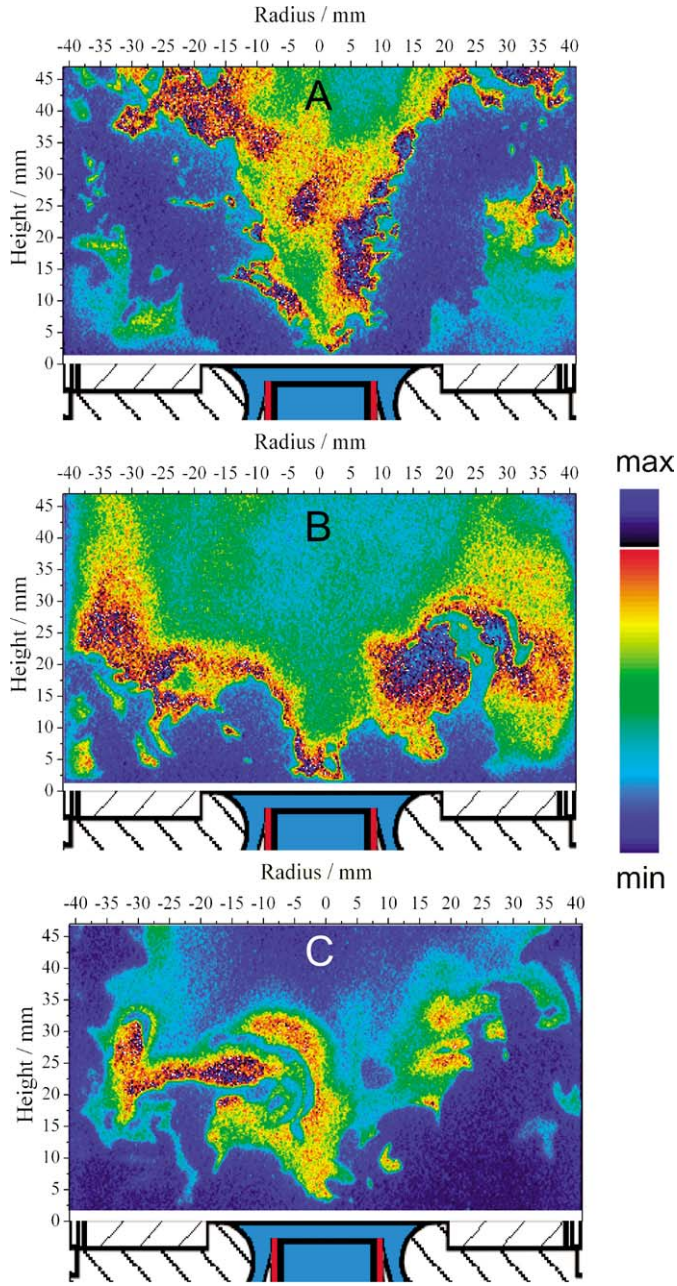

Fig. 10. Single-shot OH LIF images of flames A, B, and C.

0.072, 0.059, and 0.056 for flames A, B, and C, respectively. This demonstrates the fast mixing resulting from this nozzle configuration. In comparison, the stoichiometric mixture fraction is $f_{\text{stoich}} = 0.055$ and the overall mixture fractions of the flames were $f_{\text{glob}} = 0.037$ (A), 0.042 (B), and 0.031 (C) (see Table 1). For flames A and C, mixing is complete at $h \approx 40$ mm, and for flame B, already at $h \approx 20$ mm. These values are in agreement with the heights of the CH distributions (see Fig. 11 right), indicating that mixing and the main flame reactions are closely linked at these heights. It is further seen that in the near field of the nozzle, f is considerably higher in the irz ($f > f_{\text{glob}}$) than in the orz where $f \approx f_{\text{glob}}$. The relatively high f values in the irz enhance the stabilizing effect of the irz on the flame, because they enable a temperature level that is higher than $T_{\text{glob ad}}$. The rms values of f (displayed in Fig. 13 left) exhibit

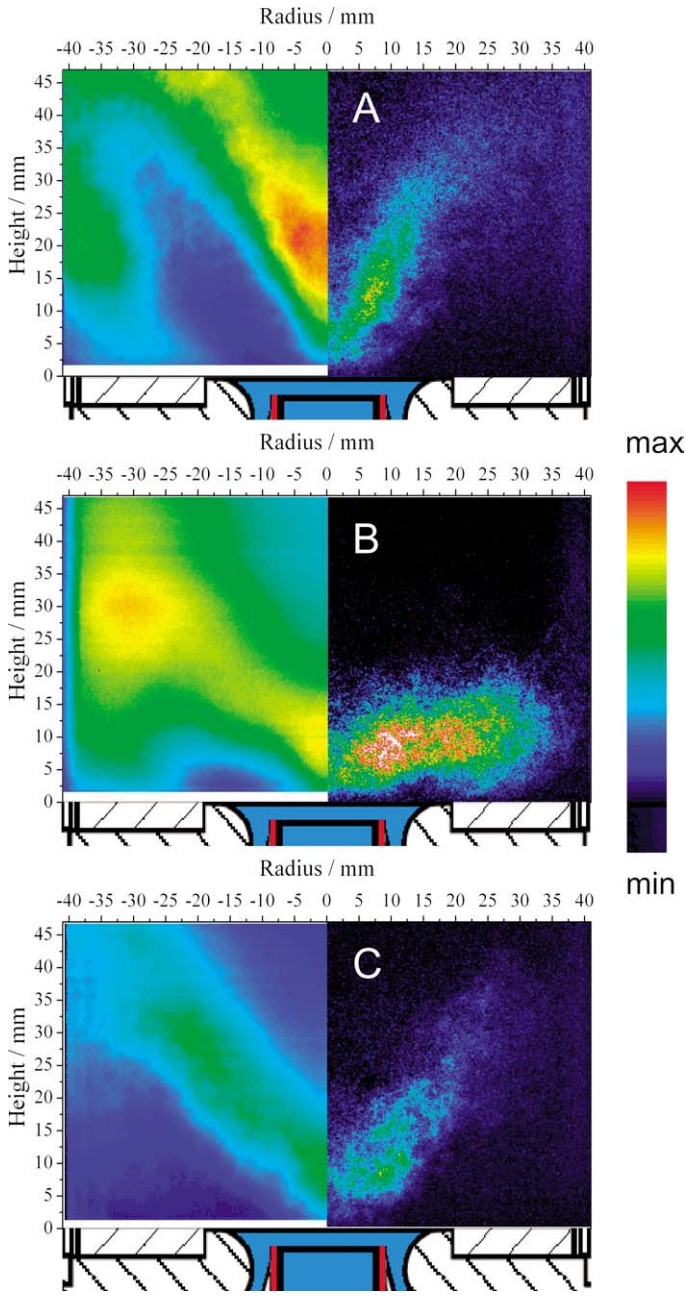

Fig. 11. Averaged OH LIF images (left) and CH LIF images (right).

a maximum of $f_{\mathrm{rms}} \approx 0.04$ at $h = 5$ mm and decrease rapidly with height. For a closer look at the mixing behavior of the three flames, Fig. 14 shows the radial profiles of f_{mean} at $h = 5$ and $h = 10$ mm. Flame A reaches the largest f_{mean} ($r = 6$ mm at $h = 5$ mm, $r = 8$ mm at $h = 10$ mm) of all three flames, which is explained by the high exit velocities of this flame and, thus, the shorter time for mixing before reaching $h = 10$ mm. It is surprising that the radial profile of f_{mean} of flame B shows a significantly smaller variation than that of flame C, although the mean flow fields are quite similar. The reason lies in the thermoacoustic oscillations of flame B: In addition to the turbulent fluctuations, the periodic variations also contribute to a homogenization of the time-averaged mixture fraction distribution.

The distributions of the mean temperatures, displayed in Fig. 15, reflect the different shapes of the

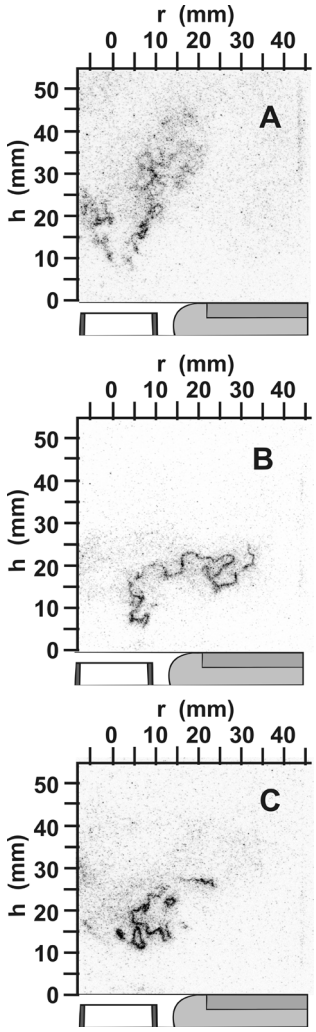

Fig. 12. Single-shot images of CH LIF.

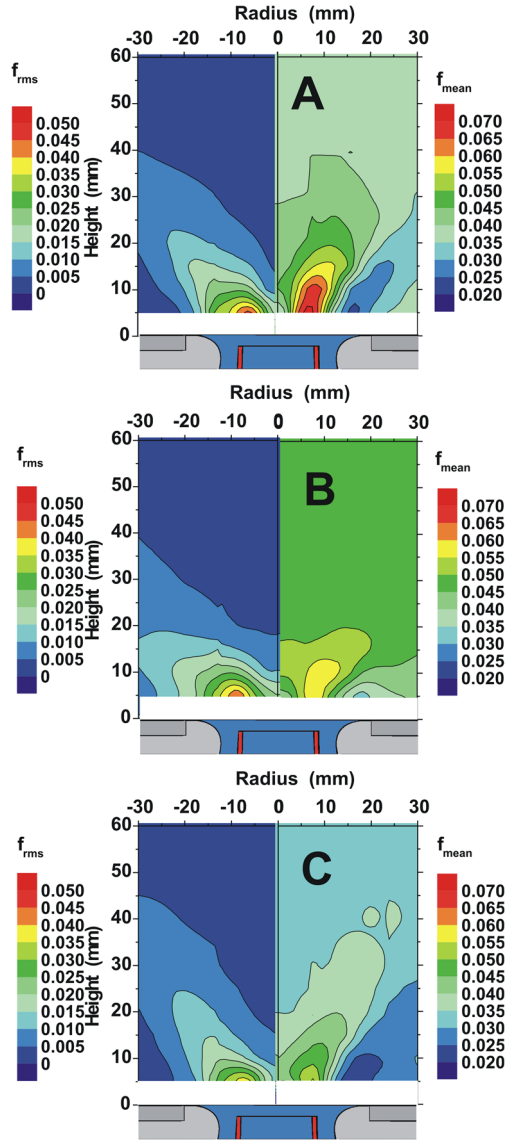

Fig. 13. Two-dimensional mixture fraction distribution (right side: mean values, left side: rms values).

flames, and it once more becomes evident that flame B is very different from the others. It is also seen that the final temperatures of the flames (at large heights) are increasing with their global equivalence ratios, as expected. To identify the differences more clearly, Fig. 16 displays the axial profiles of T_{mean} and T_{rms}. At $h = 5$ mm, all three flames exhibit a similar temperature of $T \approx 1300$ K. With increasing height, T_{mean} values increase strongly, reach a maximum at $h = 10$–20 mm, and decrease slowly afterward. For flames A and B, the maximum mean temperatures are even higher than $T_{\text{glob ad}}$ due to the relatively high f values in this region. The temperature fluctuations reach a level of 500–600 K close to the nozzle and decrease to 40–70 K at larger heights, which corresponds to the inherent rms of typically 3% due to measurement precision. Considering the temperature fluctuations within the irz, especially in the lower part, it becomes obvious that the irz is not a stationary vortex stable in time and space but, rather, is subject to significant turbulent fluctuations, as was already indicated by the single-shot images of OH and CH in this region and by the velocity fluctuations. A similar result was obtained by Ji and Gore [59] in a different swirl flame, where they showed by particle image velocimetry that the instantaneous structure of the irz is often composed of a number of smaller vortices. This must be kept in mind for the phenomenological understanding of flame behavior.

More details of the comparison between the three flames are seen in the radial profiles of T_{mean} at

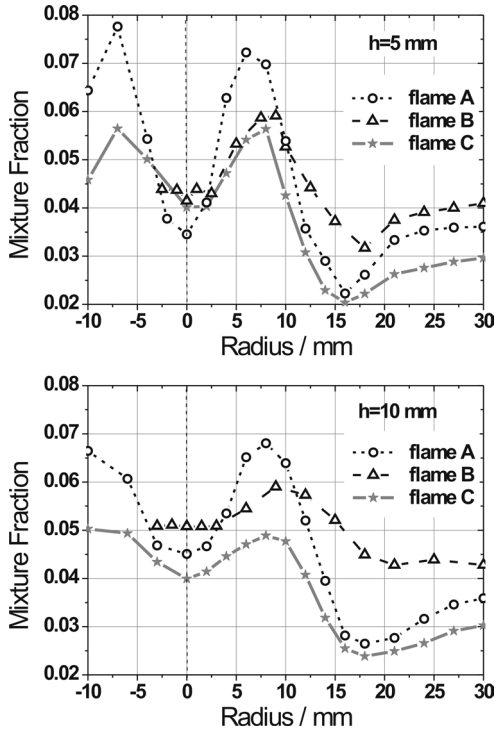

Fig. 14. Radial profiles of mean mixture fraction at $h = 5$ mm and $h = 10$ mm.

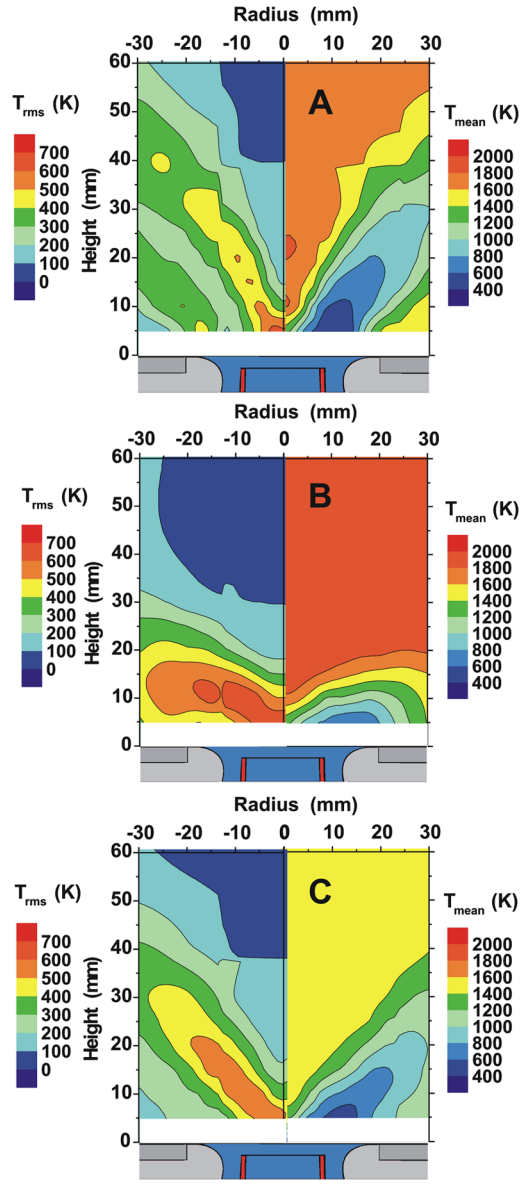

Fig. 15. Two-dimensional temperature distribution (right side: mean values, left side: rms values).

$h = 5$ and 10 mm in Fig. 17. The low-temperature regions at $r \approx 6$–17 mm reflect the inlet streams of fresh gas. Here it is seen that flame B reaches significantly higher temperatures than the other two flames; e.g., at $h = 5$ mm the lowest mean temperatures are $T_{\min} = 408$, 627, and 476 K for flames A, B, and C, respectively. The higher temperatures of flame B in this region are partly explained by the more frequent occurrence of reactions (see CH distribution in Fig. 11). However, at $h = 5$ mm and especially for $r > 10$ mm, the main source of the elevated temperatures is mixing of hot exhaust gas from the recirculation zones with fresh gas. The increased temperature level at $h = 5$ mm enhances, of course, the reactivity of the gas mixtures and, thus, the heat release and burnout [60]. This can clearly be seen in the temperature profiles at $h = 10$ mm, where flame B already reaches a minimum temperature of 1007 K, whereas for flames A and C the minimum temperatures are 554 and 639 K, respectively. The transition between the inlet stream and the orz at $r \approx 20$ mm is clearly visible in the profile of flame A at $h = 5$ mm. It is also obvious that for $h \geqslant 10$ mm, the temperature level in the orz is in general lower than in the irz, which is due to the leaner mixtures (lower f values) and heat loss to the wall in the orz.

Phase-correlated measurements in flame B revealed that the phase-resolved mean temperature of the inflowing gas at $h = 5$ mm varied by about 300 K during an oscillation cycle [34]. This variation was correlated with a periodic expansion of the recirculation zones: When the irz penetrated into the central air nozzle and the orz reached its maximum expansion, large amounts of recirculating exhaust gas were mixed into the fresh gas, increasing the temperature within the inflow. The measurements further indicated that variations of the temperature level of the inflow and the heat release rate were correlated, leading to the conclusion that the temperature of the inflow had a significant influence on the heat release rate.

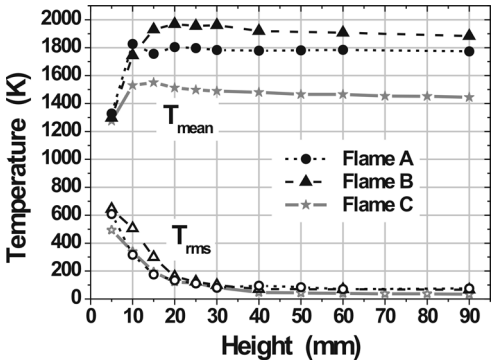

Fig. 16. Axial profiles of temperature (mean + rms).

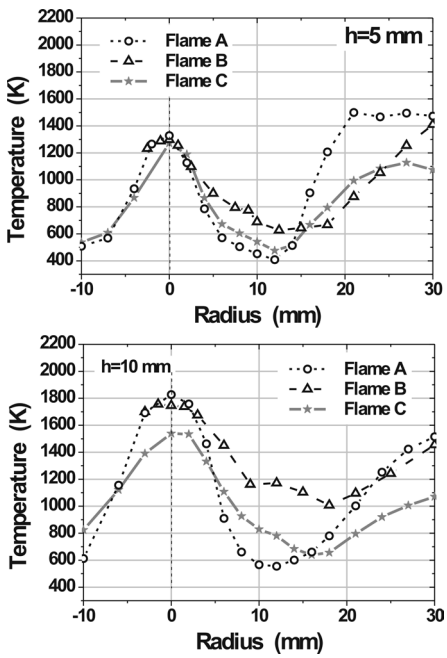

Fig. 17. Radial profiles of mean temperature at $h = 5$ mm and $h = 10$ mm.

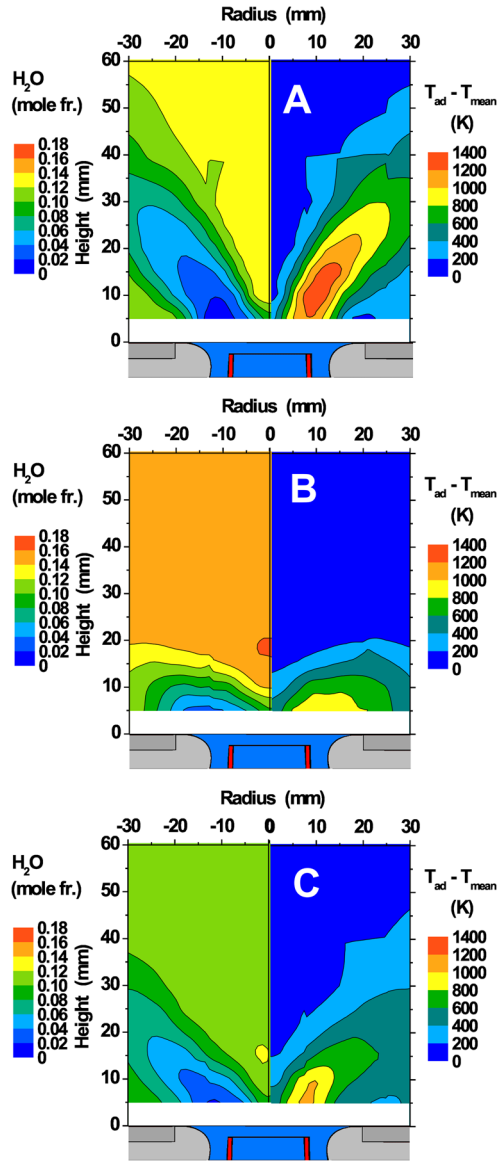

Fig. 18. Right: two-dimensional distribution of the difference between locally possible adiabatic temperature T_a and the measured mean temperature T. Left: two-dimensional distribution of the mean H_2O mole fraction. Both values can be taken as a measure of the reaction progress.

The right part of Fig. 18 displays the distributions of the mean values of the differences between the "quasi-adiabatic" flame temperature T_a and the measured temperature T. "*Quasi-adiabatic*" *flame temperature* is defined here as the temperature for the particular mixture fraction taken from a calculation for a strained laminar counterflow diffusion flame with a strain rate of $a = 1\ s^{-1}$ [53,54]. T_a has been calculated for the measured mixture fraction at each location for each single shot and from these results $T_a - T$ has been averaged. The use of the real adiabatic flame temperature (and composition) is not meaningful for fuel-rich regions of turbulent flames, because the thermal decomposition of CH_4 (which is complete for adiabatic equilibrium) takes a longer time than is typically available in these flames. Deviations between T and T_a can stem either from heat loss of the flame gases, e.g., due to thermal radiation or wall contact, or from finite-rate chemistry effects. As long as heat loss is of minor importance, the mean value of $T_a - T$ can be taken as a measure of the mean reaction progress in the flame, and is an indirect way to display the effects of finite-rate chemistry, which is discussed in more detail in the accompanying article [32]. The results displayed in

Fig. 18 show that $T_a - T$ reaches significant values in all three flames; e.g., the maximum mean values are >1400, >900, and >1000 K for flames A, B, and C just above the nozzle exit. From the large values of $T_a - T$ and its distributions, it becomes obvious that finite-rate chemistry effects play a very important role in the flames investigated. The burned gases reach a final state with $T_a - T < 200$ K that is close to equilibrium. Comparison of the results reveals that nonequilibrium effects are quite differently distributed in the three flames: Flame A reaches a constant level of $T_a - T$ at $h \approx 55$ mm, and in flame C significant effects of finite-rate chemistry are observed up to $h \approx 45$ mm, whereas in flame B a uniform $T_a - T$ is attained by $h \approx 20$ mm. These heights are in good accordance with the CH LIF images, where for flames B and C, the same heights are found for detectable CH, and for flame A, CH is still present at the upper end of the image at $h = 47$ mm. The observed difference in height between flames A and C is in accordance with the different flow velocities and Reynolds numbers of the flames. The much faster burnout of flame B is again related to the thermoacoustic pulsations and is probably caused mainly by the relatively high temperature level of the gas in the inflow as discussed before.

Finally, the mean distributions of the mole fractions X of H_2O, CH_4, and O_2 are presented in Figs. 18 and 19. The shapes of the distributions of $X(H_2O)$, displayed in Fig. 18 (left), resemble strongly those of temperature for each flame (see Fig. 15). Inspection of the single-shot results (not displayed) reveals that the correlation between $X(H_2O)$ and T is in quite good agreement with the correlations calculated for strained laminar flames. From the single-shot correlations it is, however, seen that the flames experience a temperature loss in the orz, probably due to heat conduction to the burner plate [32].

As shown in Fig. 19 (left), the distributions of $X(CH_4)$ exhibit the highest values close to the fuel nozzle; however, for flames A and C the maxima are not exactly above the CH_4 injection but shifted slightly inward. Close to the nozzle ($h < 15$ mm), the CH_4 distribution of flame B is significantly broader than those of flames A and C. A similar trend was already seen in the mixture fraction distributions (Figs. 12 and 13) and can also be observed for the O_2 distribution (Fig. 19). This result is a further indication that the periodic oscillations of the flow field generate additional mixing of fuel, air, and exhaust gas, which promotes reaction progress. The consumption of CH_4 with increasing distance from the nozzle is in general accordance with the decrease in $T_a - T$ except for a small discrepancy in the orz, where $T_a - T$ increases, due to the above-mentioned temperature loss. This can be seen comparing Fig. 18 (right) and Fig. 19 (left) at positions $r > 25$ mm and $h = 0$–10 mm. Here, the CH_4 concentration is around or smaller than 1% but the temperature difference $T_a - T$ is larger than 200 K for flame A or even 400 K for flames B and C. The temperature difference is not explainable by the remaining fuel and is probably due to heat transfer to the casing. In the irz, above $h \approx 10$ mm, and in the exhaust gas region, no CH_4 is found.

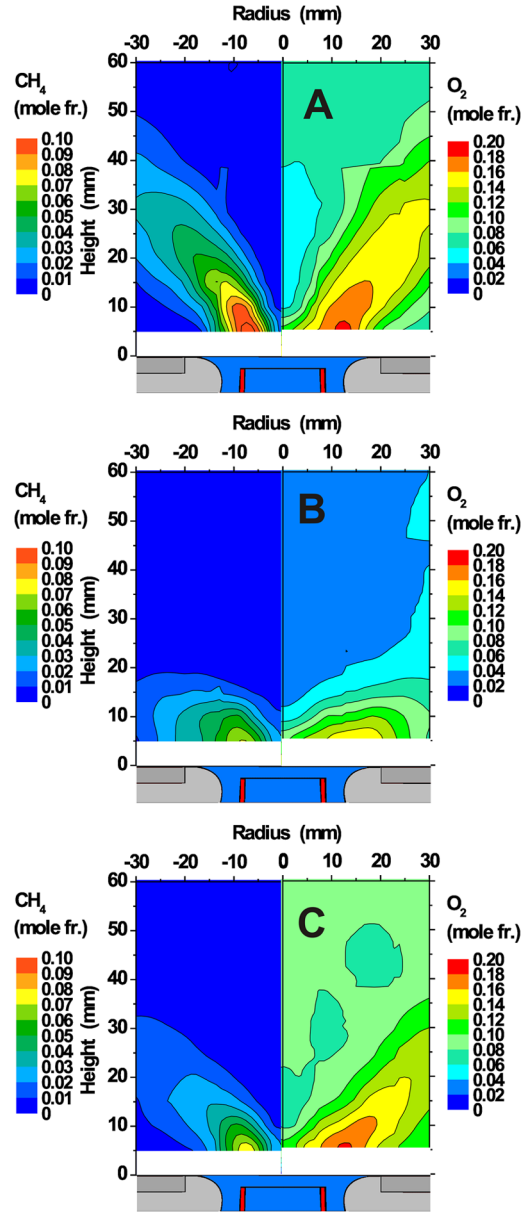

Fig. 19. Left: two-dimensional distribution of the mean CH_4 mole fraction. Right: two-dimensional distribution of the mean O_2 mole fraction.

The distributions of $X(O_2)$ (Fig. 19, right) reflect again the shapes of the flames and are in good agreement with those of T and $T_a - T$; i.e., $X(O_2)$ decreases as T increases. The fact that the lowest mean concentrations of O_2 are found near the flame axis within the irz confirms that the mixing characteristics of the burner configuration generate a relatively fuel-rich (but still overall lean) recirculating gas flow which is of importance for flame stabilization. The distributions of the remaining major species (N_2, CO_2, CO, H_2) are in good agreement with the results presented and are not displayed. The mean mole fractions of the intermediate species CO and H_2 are generally below 0.018 and 0.012, respectively. The highest concentrations of these species are found in the shear layer between the inlet flow and the irz at $h = 10$–20 mm with near-stoichiometric mixture fractions and temperatures between 1000 and 1500 K.

4. Summary and conclusions

A laboratory-scale GT model combustor has been described and three CH_4/air diffusion flames with different flame characteristics have been investigated using LDV, PLIF of OH and CH, and laser Raman scattering. The main goals of the studies were (1) to carry out a detailed experimental analysis to improve understanding of the physical and chemical processes leading to the different behaviors of swirling flames, and (2) to provide a useful database for the verification and improvement of numerical combustion models. The results presented in this article concern the flow fields and the flame structures, as well as the mean values and fluctuations of the major species concentrations, mixture fraction, and temperature. A second article addresses turbulence–chemistry interactions and their effect on mixing and stabilization [32].

The shapes of the mean flow fields of all three flames are quite similar: The injected flows of CH_4 and air formed a cone with an opening angle $\theta/2$ of about 26° with respect to the flame axis. Due to the swirl, a pronounced irz was established which reached from $h \approx 70$ mm down into the central air nozzle. This reverse flow of hot combustion products formed the major source for the ignition and stabilization of the flames. The instantaneous flame structures and velocity fluctuations gave evidence that the irz was, at least in the lower part, subject to strong turbulent fluctuations of its shape and composition. An orz was established in the lower part of the combustion chamber.

The flame structures were visualized by planar laser-induced fluorescence of OH and CH. The results showed that the instantaneous flame shapes were dominated by turbulence and that the three flames could hardly be distinguished based on a single instantaneous OH or CH image. The reaction zones of all three flames were generally thin ($\leqslant 0.5$ mm) and strongly wrinkled but more contorted in flame A, which had a higher Reynolds number. The averaged images of CH PLIF showed the regions of heat release. It could be seen that none of the flames was attached to the nozzle; i.e., reactions did not start below $h \approx 5$, 4, and 6 in flames A, B, and C, respectively. The flames were thus partially premixed before ignition. Flame B had a remarkably short flame length of $h \approx 20$ mm, whereas the other two flames reached up to $h \approx 50$ mm (flame A) and $h \approx 40$ mm (flame C), which also represented a fast burnout. The averaged OH and CH distributions also revealed the significantly different shapes of the three flames. The opening angles for the flame zones deviated from the opening angle of the flow fields for flame C and especially for flame B, while for flame A the two angles matched roughly.

While a certain similarity was seen for flames A and C, with opening angles $\theta/2$ of 30° and 45°, respectively, flame B had a significantly larger opening angle ($\theta/2 = 75°$) and was much shorter. The different shapes of flames B and C were surprising because their mean flow fields were very similar, especially at the nozzle exit. Except for the velocity fields, all other measured quantities confirmed that flame B exhibited a different behavior than flames A and C. Microphone measurements revealed that the sound emissions of flame B were concentrated at a frequency of ≈290 Hz, proving the occurrence of thermoacoustic oscillations, and previously reported phase-resolved LDV and PLIF measurements showed significant periodic variations of the flame structure during an oscillation cycle. Periodic changes in the expansion of the inner and outer recirculation zones enhanced the mixing of fuel, air, and exhaust gas, which in turn contributed to increased reaction progress.

Measurements of the mixture fraction demonstrated the fast mixing of fuel and air of this nozzle configuration. At $h = 5$ mm, variations of the mean mixture fraction along the radial profile remained between $f_{mean} = 0.02$ and 0.08 for flame A and were even smaller for the other flames. Mixing was completed at about the same height as the heat release, i.e., at $h \approx 40$ mm for flames A and C and at $h \approx 20$ mm for flame B. Within the irz, f and T were higher and $X(O_2)$ was smaller than the global values. The flow and the mixing characteristics of the burner enhanced the effect of the irz with respect to ignition and stabilization of the flame. Although measurements could not be performed below $h = 0$ mm, the results obtained indicated that hot combustion products were transported via the irz into the central nozzle, where

they mixed with air before reentering the combustion chamber.

The distributions of $T_a - T$ showed that in all three flames, pronounced finite-rate chemistry effects occurred which led to a significant deviation from equilibrium composition and temperature, especially within the high-velocity regions of the inlet flow and the neighboring shear layers. Finally, it should be noted that the experimental data are available from one of the authors (W.M.) on request for validation of numerical codes for swirling turbulent flames.

Acknowledgments

The work presented here was performed mainly in the frame of the project Combustion Control and Simulation, funded by the State of Baden-Württemberg, and as part of the DLR project NACOS. The financial support within these projects is gratefully acknowledged by the authors. We furthermore thank B. Lehmann for execution of the LDV measurements and B. Noll for fruitful discussions.

References

[1] A.K. Gupta, D.G. Lilley, N. Syred, Swirl Flows, Abacus Press, Kent, 1984.
[2] N. Syred, N.A. Chigier, J.M. Beér, Proc. Combust. Inst. 13 (1971) 617–624.
[3] N. Syred, J.M. Beér, Combust. Flame 23 (1974) 143–201.
[4] R. Weber, J. Dugué, Prog. Energy Combust. Sci. 18 (1992) 349–367.
[5] S.M. Correa, Proc. Combust. Inst. 27 (1998) 1793–1807.
[6] A.H. Lefebvre, Gas Turbine Combustion, Taylor & Francis, Philadelphia, 1999.
[7] H.J. Bauer, Prog. Comput. Fluid Dyn. 4 (2004) 130–142.
[8] W. Meier, X.R. Duan, P. Weigand, B. Lehmann, Gaswärme Int. 53 (2004) 153–158.
[9] W. Stricker, in: K. Kohse-Höinghaus, J. Jeffries (Eds.), Applied Combustion Diagnostics, Taylor & Francis, New York, 2002, pp. 155–193.
[10] A.C. Eckbreth, Laser Diagnostic for Combustion Temperature and Species, Gordon & Breach, 1996.
[11] K. Kohse-Höinghaus, J. Jeffries (Eds.), Applied Combustion Diagnostics, Taylor & Francis, New York, 2002.
[12] A.R. Masri, R.W. Dibble, R.S. Barlow, Prog. Energy Combust. Sci. 22 (1996) 307–362.
[13] J. Wolfrum, Proc. Combust. Inst. 28 (1998) 1–41.
[14] H. Kaaling, R. Ryden, Y. Bouchie, D. Ansart, P. Magre, C. Guin, in: 13th International Symposium on Air Breathing Engines (ISABE), Chattanooga, TN (USA), 1997.
[15] S. Kampmann, T. Seeger, A. Leipertz, Appl. Opt. 34 (1995) 2780–2786.
[16] F. Dinkelacker, A. Soika, D. Most, D. Hofmann, A. Leipertz, W. Polifke, K. Döbbeling, Proc. Combust. Inst. 27 (1998) 857–865.
[17] R. Fink, A. Hupfer, D. Rist, in: Proceedings, ASME Turbo Expo 2002, GT-2002-30078.
[18] C.S. Cooper, N.M. Laurendeau, Appl. Phys. B 70 (2000) 903–910.
[19] C.S. Cooper, N.M. Laurendeau, Combust. Flame 123 (2000) 175–188.
[20] W.-P. Shih, J.G. Lee, D.A. Santavicca, Proc. Combust. Inst. 26 (1996) 2771–2778.
[21] Y. Deguchi, M. Noda, Y. Fukuda, Y. Ichinose, Y. Endo, M. Inida, Y. Abe, S. Iwasaki, Meas. Sci. Technol. 13 (2002) R103–R115.
[22] P.O. Hedman, D.L. Warren, Combust. Flame 100 (1995) 185–192.
[23] S.-Y. Lee, S. Seo, J.C. Broda, S. Pal, R.J. Santoro, Proc. Combust. Inst. 28 (2000) 775–782.
[24] A. Arnold, R. Bombach, W. Hubschmid, B. Käppeli, ERCOFTAC Bull. 38 (1998) 10–19.
[25] J. Fritz, M. Kröner, Th. Sattelmayer, in: Proceedings, ASME Turbo Expo 2001, 2001-GT-0054.
[26] C. Löfström, J. Engström, M. Richter, C.F. Kaminsky, P. Johansson, K. Nyholm, J. Nygren, M. Aldén, in: Proceedings, ASME Turbo Expo 2000, 2000-GT-0124.
[27] C.M. Gittins, S.U. Shenoy, H.R. Aldag, D.P. Pacheco, M.F. Miller, M.G. Allen, in: 38th AIAA Aerospace Sciences Meeting, Reno, NV, 2000.
[28] U.E. Meier, D. Wolff-Gaßmann, J. Heinze, M. Frodermann, I. Magnusson, G. Josefsson, in: 18th International Congress on Instrumentation in Aerospace Simulation Facilities (ICIASF 99), Toulouse, 1999, pp. 7.1–7.7.
[29] U.E. Meier, D. Wolff-Gaßmann, W. Stricker, Aerospace Sci. Technol. 4 (2000) 403–414.
[30] M. Carl, T. Behrendt, C. Fleing, M. Frodermann, J. Heinze, C. Hassa, U. Meier, D. Wolff-Gaßmann, S. Hohmann, N. Zarzalis, ASME J. Eng. Gas Turbines Power 123 (2001) 810–816.
[31] O. Kunz, B. Noll, R. Lückerath, M. Aigner, S. Hohmann, in: 37th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibition, Salt Lake City, UT, 2001, AIAA 2001-3706.
[32] W. Meier, X. Duan, P. Weigand, Combust. Flame 144 (2006) 225–236.
[33] R. Giezendanner, O. Keck, P. Weigand, W. Meier, U. Meier, W. Stricker, M. Aigner, Combust. Sci. Technol. 175 (2003) 721–741.
[34] X.R. Duan, W. Meier, P. Weigand, B. Lehmann, Appl. Phys. B 80 (2005) 389–396.
[35] P. Weigand, W. Meier, X.R. Duan, R. Giezendanner-Thoben, U. Meier, Flow Turbulence Combust., in press.
[36] M. Cao, H. Eickhoff, F. Joos, B. Simon, in: ASME Propulsion and Energetics, 70th Symposium, AGARD Conf. Proc. 422 (1987) 8.1.
[37] W. Leuckel, N. Fricker, J. Inst. Fuel 49 (1976) 103–112.
[38] D.G. Lilley, AIAA J. 15 (8) (1977) 1063–1078.

[39] J.A. Sutton, J.F. Driscoll, Appl. Opt. 42 (2003) 2819–2828.
[40] J. Tobai, T. Dreier, J.W. Daily, J. Chem. Phys. 116 (2002) 4030–4038.
[41] G.H. Dieke, H.M. Crosswhite, J. Quant. Spectrosc. Radiat. Transfer 2 (1962) 97–199.
[42] J. Luque, D.R. Crosley, J. Chem. Phys. 104 (1996) 3907–3913.
[43] J. Luque, D.R. Crosley, SRI Report No. MP 98-021, SRI Int., Menlo Park, CA, 1998.
[44] C.D. Carter, J.M. Donbar, J.F. Driscoll, Appl. Phys. B 66 (1998) 129–132.
[45] J. Luque, R.J.H. Klein-Douwel, J.B. Jeffries, D.R. Crosley, Appl. Phys. B 71 (2000) 85–94.
[46] O. Keck, W. Meier, W. Stricker, M. Aigner, Combust. Sci. Technol. 174 (2002) 73–107.
[47] P. Weigand, R. Lückerath, W. Meier, http://www.dlr.de/VT/Datenarchiv, 2003.
[48] V. Bergmann, W. Meier, D. Wolff, W. Stricker, Appl. Phys. B 66 (1998) 489–502.
[49] G. Dixon-Lewis, Proc. Combust. Inst. 23 (1990) 305–324.
[50] R.S. Barlow, R.W. Dibble, J.-Y. Chen, R.P. Lucht, Combust. Flame 82 (1990) 235–251.
[51] J. Luque, R.J.H. Klein-Douwel, J.B. Jeffries, G.P. Smith, D.R. Crosley, Appl. Phys. B 75 (2002) 779–790.
[52] H.N. Najm, P.H. Paul, C.J. Mueller, P.S. Wyckoff, Combust. Flame 113 (1998) 312–332.
[53] J.-Y. Chen, University of California at Berkeley, private communication.
[54] J.H. Miller, R.J. Kee, M.D. Smooke, J.F. Grcar, Paper WSS/CI 84-10, in: Western State Section of the Combustion Institute, Spring Meeting, 1984.
[55] U. Rahmann, A. Bütler, U. Lenhard, R. Düsing, D. Markus, A. Brockhinke, K. Kohse-Höinghaus, LASKIN—A Simulation Program for Time-Resolved LIF-Spectra, International Report, University of Bielefeld, Faculty of Chemistry, Physical Chemistry I, http://pc1.uni-bielefeld.de/laskin V. 5.0.
[56] K.J. Rensberger, M.J. Dyer, R.A. Copeland, Appl. Opt. 27 (1988) 3679–3689.
[57] R. Giezendanner-Thoben, U. Meier, W. Meier, M. Aigner, Flow Turbulence Combust., in press.
[58] W. Meier, X.R. Duan, P. Weigand, Proc. Combust. Inst. 30 (2005) 835–842.
[59] J. Ji, J.P. Gore, Proc. Combust. Inst. 29 (2002) 861–867.
[60] M. Braun-Unkhoff, DLR Stuttgart, private communication.